Rachid Elfermi

Bioinformatics Engineer | Human & Environmental Genomics

Morocco · Current research role in Rabat · Updated 12 October 2026

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Bioinformatics engineer and computational biology researcher with approximately five years of environmental genomics and metagenomics research at UM6P. Current work spans human WES/WGS, germline variant calling, variant annotation and prioritization, pipeline development and health-related microbiome research.

PROFESSIONAL EXPERIENCE

Bioinformatician

August 2026 - present | Hub de Médecine de Précision (HMP)

Specialized precision medicine unit within the Hôpital Universitaire International Mohammed VI de Rabat, part of the Fondation Mohammed VI des Sciences et de la Santé (FM6SS) ecosystem.

• Human whole-exome sequencing (WES) and whole-genome sequencing (WGS) analysis.
• Germline variant calling, annotation, filtering and prioritization.
• Bioinformatics pipeline development and human-health microbiome research.

Bio-informaticien - Environmental genomics

December 2021 - August 2026 | Mohammed VI Polytechnic University (UM6P)

Approximately five years of research in environmental genomics, metagenomics and microbial bioinformatics.

• Developed genomic and metagenomic computational workflows.
• Taxonomic, functional and ecological profiling of environmental and host-associated microbial communities.
• Supported students and researchers through bioinformatics training.

Bioinformatics internship - Metagenomic gene prediction

2020 | UM6P, Benguerir

Developed an ORF finder for complete and incomplete metagenomic fragments, ten convolutional neural network models across GC-content ranges, and a configurable integration script. This internship is distinct from the later five-year research chapter.

Software Engineering

January - June 2018 | Domaine agricole ELFERMI, Ait Ali

Designed, implemented and maintained a JavaFX human resources application.

Technology intern

November - December 2017 | Atlasblessings, Agadir

Team-based website development and maintenance using BigCommerce CMS and Toukan app services.

Scientific toolkit & research outputs

Human genomics: WES/WGS, germline variant calling, annotation, filtering, prioritization, genomic data interpretation and health-related microbiome research.

Environmental genomics: quality control (FastQC, Sickle), assembly (SPAdes), annotation (Prokka), pan-genomes (BPGA, Roary), phylogenetics (MEGA X, CLC Genomics Workbench), 16S rRNA profiling.

Programming & computing: Python, R, Bash, Linux, Java, C, C#, SQL, HTML/CSS/JavaScript/PHP, statistics, machine learning, CNNs, UML and Merise.

EDUCATION

Master's degree in Bioinformatics | Faculty of Sciences of Meknes | 2018 - 2021
Bachelor's degree in Software Engineering | Polydisciplinary Faculty of Ouarzazate | 2013 - 2017

Master's dates follow the current LinkedIn profile; the historical CV gives 2020 as the end year. The internship is listed as 2020 because exact months differ between sources. Continuing learning: OpenClassrooms Python courses (2019).

SELECTED CO-AUTHORED PUBLICATIONS

Plant identity and environmental filtering are the key drivers of bacterial community structure in four desert plant species from the Sahara Desert in Morocco

BMC Microbiology · 25, 734 | 2025
Environmental microbiome · Software

Cervico-Vaginal Microbiome Dynamics Across HPV-Driven Lesion Stages in Moroccan Women

Microorganisms · 13(8), 1884 | 2025
Human-health microbiome · Formal analysis

HPV genotypes in Africa: comprehensive analysis of genetic diversity and evolutionary dynamics

Archives of Virology · 170, 116 | 2025
Human health · Viral genomics · Formal analysis, validation & visualization

Risk factors for cervical cancer in Morocco: a case-control study

Pan African Medical Journal · 50, 87 | 2025
Human health research · Review & validation

Enhancing chickpea growth through arbuscular mycorrhizal fungus inoculation: facilitating nutrient uptake and shifting potential pathogenic fungal communities

Mycorrhiza · 35, 1 | 2025
Environmental microbiome · Methodology, data curation & review

Afro-TB dataset as a large scale genomic data of Mycobacterium tuberuclosis in Africa

Scientific Data · 10, 212 | 2023
Microbial genomics · Co-authored data descriptor

RESEARCH COLLABORATION

Open to scientific discussions with Moroccan and international teams in genomics, microbiome research, metagenomics and computational biology. General methods, including shotgun metagenomics, are explained in the portfolio; project-specific tool stacks and outcomes are described only where documented.